HR Policy Assistant

Project rating and improvement list | 5 November 2026

Repository: github.com/Rutuja-Murkute98/hr-policy-assistant

Portfolio or learning project	**8.5 / 10**	It has a clean offline/online split, grounded answers, measured quality and offline tests.
Production readiness	**5 / 10**	It has no user login, chat memory, automated re-indexing, monitoring or CI, and the hosting work isn't committed yet.

These scores are my judgement, based on the code I read and the checks I ran. The strongest parts are the engineering habits, such as measuring quality and refusing when unsure. The weakest part is that it only handles one fixed set of documents for anonymous users.

Tools and technologies

Layer	Tool	Score	Assessment
Chunking	Custom heading splitter	**9**	This is the best design decision in the project, because each chunk is one section and keeps its heading path.
Vector database	PostgreSQL + pgvector	**9**	It's a practical choice with no separate vector service. It has no vector index yet, which is fine at 539 chunks.
Backend	FastAPI	**9**	The API is small and clean, with typed models and safe errors. It lacks streaming and API versioning.
Testing	pytest with fakes	**8.5**	The tests are fast and use no quota. There's no CI, and no test runs against a real database.
Retrieval	MMR + distance limit	**8**	It's well tuned and explained, but it's vector-only and the limit has a thin margin.
Chat model	Gemini Flash Lite	**8**	It suits a free-tier project, but temperature isn't set and the provider is hard-coded.
Embeddings	Gemini embeddings	**8**	They work well. At 3,072 dimensions, pgvector's `vector` type can't use its standard indexes, which stop at 2,000.
Evaluation	RAGAS + custom checks	**7.5**	This is rare in projects at this level. The test set is small, and the same model writes and judges the answers.
Framework	LangChain	**7**	It's used lightly, which keeps the code readable. Dependencies are only partly pinned, and RAGAS needs a workaround.
Local database	pixeltable-pgserver	**7**	It's a clever way to avoid Docker, but it's only for local use.
Frontend	Streamlit + custom CSS	**7**	It's quick and polished, but the CSS is tied to Streamlit's internals and it won't suit a full product.
Security	API key + in-memory limiter	**6**	It's fine for one process, but it can't identify users and its counts reset on restart.
Deployment	Render + Streamlit Cloud + hosted Postgres	**6**	The design is sensible for free hosting, but it's uncommitted and `DEPLOY.md` is missing.
Monitoring	Standard logging	**5**	You get request IDs and timings, but no token, cost or retrieval tracking.

Improvements

Effort is S for hours, M for days and L for a week or more.

Conversation

Improvement	What it adds	Effort
Chat memory with question rewriting	Follow-ups like "what about on probation?" work	**M**
Real streaming over SSE	Answers appear as they're generated, replacing the animation	**M**
Clarifying questions	The bot asks "which band?" instead of giving a vague answer	**M**
Thumbs up/down feedback	Gives you real data on answer quality	**S**
Hindi and Marathi support	Employees can ask and get answers in their own language	**M**

Knowledge base

Improvement	What it adds	Effort
Change detection with content hashes	Re-embeds only edited sections and removes deleted ones	**S**
More file types (PDF, DOCX, HTML)	Works with documents as companies actually store them	**M**
Admin upload screen	HR can update policies without a developer	**M**
Connectors (SharePoint, Google Drive, Confluence)	Keeps the index in sync automatically	**L**
Keep version and effective dates	Answers can state which version applies, since the chunker currently drops these lines	**S**

Retrieval

Improvement	What it adds	Effort
Hybrid search (keyword + vector)	Finds exact terms like "FIR", "PIP" or "ESOP" more reliably	**M**
Reranker	Puts the most useful chunks first	**M**
Policy routing with metadata filters	Narrows the search to the right policy, as your guide's missing `router.py` intended	**M**
Glossary-based query expansion	Connects abbreviations and HR terms to the policy wording	**S**
Parent-section retrieval	Sends the full section when a split piece matches	**S**
Answer cache	Reduces cost and speeds up repeated questions	**S**
Vector index using `halfvec` or fewer dimensions	Keeps search fast as documents grow	**S**

Personalisation and actions

Improvement	What it adds	Effort
Employee login (SSO)	Identifies real users instead of just the UI server	**M**
Employee profile in answers	Answers fit the person's band, location and probation status	**M**
Document access control	Restricts HR-only or manager-only content	**M**
HRMS tools	Answers live questions like "what's my leave balance?"	**L**
Human handoff	Creates an HR ticket with the conversation attached	**M**
Multi-company support	Lets one deployment serve several organisations	**L**

Answer quality and safety

Improvement	What it adds	Effort
Inline citations with quotes	Shows exactly which text supports each claim	**M**
Structured output with a refusal flag	Replaces the fragile exact-string refusal check	**S**
Low temperature setting	Makes answers more consistent	**S**
Sensitive-topic routing	Sends POSH, grievance and whistleblower questions to the right people with care	**M**
Personal data redaction	Protects details employees type into questions	**M**
Stronger prompt injection checks	Adds defences beyond a single prompt rule	**M**

Platform and integrations

Improvement	What it adds	Effort
Swappable LLM and embedding providers	Lets you change model through config, including Claude, OpenAI or local models	**M**
Slack and Teams bots	Puts the helpdesk where employees already work, and proves the API-first design	**M**
Versioned API (`/v1`)	Lets clients upgrade safely	**S**
Redis rate limiting	Works across several server processes and survives restarts	**S**
Docker Compose	Starts the whole system with one command	**S**
CI pipeline	Runs tests and checks on every push	**S**
Tracing (Langfuse or OpenTelemetry)	Shows retrieved chunks, tokens, cost and latency for each request	**M**
Admin analytics dashboard	Shows top questions and unanswered topics, revealing gaps in the policies	**M**

Evaluation

Improvement	What it adds	Effort
Test set of 100+ questions	Covers paraphrases, multi-policy questions and attacks	**M**
Section-level hit rate	Current file-level checks are too easy, as the old chunking also scored 100%	**S**
Saved distances per question	Lets you tune the 0.42 limit from data	**S**
Independent judge model	Avoids the same model grading its own answers	**S**
Evaluation gate in CI	Blocks changes that make answers worse	**M**

Housekeeping

Improvement	What it adds	Effort
Commit the hosting work and write `DEPLOY.md`	Makes the deployment reproducible	**S**
Fix outdated docs	Corrects the README test count and the guide's missing files	**S**
Pin all dependencies	Stops upgrades from breaking fresh installs	**S**

What I'd do first

1. **Finish and deploy:** commit the hosting work, add `DEPLOY.md`, then add CI. A live link adds the most to a portfolio project.
2. **Chat memory and streaming:** these are the changes users will notice most.
3. **Change detection and an admin upload screen:** policy updates stop needing a developer, which makes the project far more dynamic.
4. **Hybrid search and a reranker:** these help most on exact terms and abbreviations, then the larger test set can measure the gain.
5. **Swappable providers and a Slack or Teams bot:** these make it far more versatile, and show the separate API was worth building.